

Ferric Lithosols

Ferric Lithosols are very shallow, stony soils that form over hard rock or iron-rich weathered materials.

The term “*Ferric*” refers to the dominance of iron oxides, which impart a reddish to dark brown coloration. These soils represent the early stages of soil development and are common in upland landscapes where erosion and rock exposure limit soil formation.

They are typically found on steep slopes, rocky hills, and dissected terrains, often in semi-arid to sub-humid regions.

Drainage:

- *Wet season:* Well to excessively drained due to their shallow depth and stony nature, resulting in rapid surface runoff and minimal infiltration.
- *Dry season:* Soils dry quickly and retain little moisture. Water stress is common for plants, especially during prolonged dry periods.
- *Landscape position:* Found on upper slopes, hill crests, and rocky outcrops where erosion dominates over deposition.

Depth:

- Very shallow to shallow, generally less than 30–50 cm before reaching hard bedrock or ferric crust.
- Rooting depth is highly restricted, with only shallow-rooted vegetation able to thrive.

Workability:

- When moist, the soil is friable and can be lightly tilled.
- When dry, it becomes hard and crusty, making cultivation difficult.
- The stony and shallow nature limits mechanization; hand tools or light equipment are preferred.
- Workable only within a narrow moisture window after rainfall.

Nutrient Status & Cation Exchange Capacity (CEC):

- CEC: Very low (3–10 cmol(+)/kg) due to dominance of iron and aluminum oxides and low clay activity.
- Base saturation: Low, reflecting limited exchangeable Ca^{2+} , Mg^{2+} , and K^{+} .
- Nutrient levels: Deficient in nitrogen, phosphorus, and potassium. Organic matter low to moderate (0.5–1.5%) depending on vegetation cover.
- Micronutrients such as Fe and Mn are abundant, while Zn and B may be marginal under continuous cropping.

pH:

- Generally moderately to strongly acidic (pH 5.0–6.0).
- May approach neutrality in areas developed from basic parent rocks like basalt.

Management Recommendations:

- Maintain continuous vegetation cover to minimize erosion and surface sealing.
- Avoid intensive cultivation or deep tillage to prevent exposure of bedrock and further degradation.
- Promote agroforestry, silvopastoral systems, or perennial shallow-rooted crops such as sisal, cashew, or grasses for grazing.
- Apply organic residues, compost, or mulch to enhance moisture retention and nutrient supply.
- Where small-scale cultivation is practiced, use light implements and timed cultivation shortly after rainfall.
- Supplement nutrients through targeted fertilizer or organic inputs, emphasizing nitrogen and phosphorus.

Suitable Uses:

Primarily suited for rangeland, woodland, or low-intensity grazing. Limited crop suitability, but with proper management, may support hardy perennial crops, shrubs, or managed pasture.